

Sithija Manage

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EDUCATION

2022 - 2027	Ph.D. in Statistics Cornell University, Ithaca, NY
2019 - 2022	B.S. in Applied Mathematical Sciences Texas A&M University, College Station, TX

Relevant Coursework:

Mathematics of Deep Learning, Understanding Machine Learning, Causal Inference & Machine Learning, Bayesian Statistics & Data Science
Statistical Computing, Probability Theory, Mathematical Statistics, Multivariate Analysis

RESEARCH EXPERIENCE

2025 - Present	Visiting Graduate Student Researcher Harvard University Department of Biostatistics Advisor: Curtis Huttenhower
2023 - Present	Graduate Student Researcher Cornell University Department of Statistics and Data Science Advisors: Martin Wells, Sam Wang
2021 - 2022	Undergraduate Student Researcher Texas A&M University Department of Mathematics Advisor: Sai-Mang Pun
2021 - 2021	Undergraduate Student Researcher University of North Carolina at Greensboro Department of Mathematics and Statistics Advisor: Xiaoli Gao
2020 - 2021	Undergraduate Student Researcher Texas A&M University Department of Mathematics Advisor: Wencai Lu

HONORS AND AWARDS

2025	Cornell Graduate School Research Travel Grant - (Harvard Biostatistics) \$1,600
2024-2025	National Institutes of Health (NIH) Artificial Intelligence and Precision Nutrition T32 Training Grant - Cornell University - NIH \$50,995 (to date)
2019-2023	Presidents Endowed Scholarship – Texas A&M University
2021-2022	William A. Triche Scholarship – Texas A&M University College of Science \$45,000
2020	Dr. Walter E. Koss Endowed Scholarship - Texas A&M Department of Mathematics \$2,000
2019	National Merit Finalist
2018	Academy Mathematics Student of the Year - Selected by Board of Academy Staff
2016-2017	American Statistical Association Special Award – State-level and Regional Science Fairs

PUBLICATIONS

Peer-Reviewed Manuscripts

- [1] Sarkar, R., **Manage, S.**, and Gao, X., “Stable Variable Selection for High-Dimensional Genomic Data with Strong Correlations,” *Annals of Data Science*, Jun. 2023. DOI: <https://doi.org/10.1007/s40745-023-00481-5>.
- [2] **Manage, S.**, “Neural Network Approach to NFL Position Classification,” *SIAM Undergraduate Research Online*, vol. 15, Jul. 2022. DOI: [10.1137/21S1444485](https://doi.org/10.1137/21S1444485).
- [3] Hatinoğlu, B., Eakins, J., Frendreiss, W., **Manage, S.**, Lucille, L., and Puente, A., “Ambarzumian-type Problems for Discrete Schrödinger Operators,” *Complex Analysis and Operator Theory*, no. 15, p. 118, Oct. 2021. DOI: <https://doi.org/10.1007/s11785-021-01169-5>.

Preprints

- [4] **Manage, S.**, Wang, Y. S., and Wells, M. T., “Nonparametric regression via tree-guided feature aggregation,” *arXiv preprint arXiv:2605.26653*, 2026. arXiv: [2605.26653](https://arxiv.org/abs/2605.26653).

Manuscripts in Preparation

- [5] **Manage, S.**, Accorsi, E., Nearing, J., Ma, S., Franzosa, E., and Huttenhower, C., *Effect-size Estimation and Confounding-Aware Permutations for PERMANOVA*, Manuscript in preparation, 2026.

PRESENTATIONS

Oral

- [1] *Kernel Regression with Tree-Exploring Aggregations for Microbiome Analysis*, Harvard Medical School/Brigham and Women’s Hospital - Georg Gerber Lab, Feb. 2026.
- [2] *Effect-Size Estimation and Confounding-Aware Permutations for PERMANOVA*, Department of Biostatistics at Harvard University - Curtis Huttenhower Lab, Feb. 2026.
- [3] *Modern Statistical Tools for Microbiome Data Analysis*, Sam Houston State University Department of Mathematics and Statistics Seminar, Oct. 2025.

- [4] *Causal Inference for Microbiome Data*, Department of Biostatistics at Harvard University - Curtis Huttenhower Lab, Oct. 2025.
- [5] *Stable Variable Selection for High-Dimensional Genomic Data with Strong Correlations (3rd place - Engineering/Math Category)*, American Medical Student Association Research Symposium, 2021.

Poster

- [1] *Kernel Regression with Tree-Exploring Aggregations*, Celebration of Statistics and Data Science, Cornell University, Sep. 2025.
- [2] *Kernel Regression with Tree-Exploring Aggregations*, Joint Statistical Meetings (JSM), Aug. 2025.
- [3] *Kernel Regression with Tree-Exploring Aggregations*, Harvard Chan Microbiome in Public Health Symposium, May 2025.
- [4] *Complexities of Causal Compositional Data Analysis*, Celebration of Statistics and Data Science, Cornell University, Sep. 2024.
- [5] *Complexities of Causal Compositional Data Analysis*, Joint Statistical Meetings (JSM), Aug. 2024.
- [6] *Ambarzumian- type Problems for Discrete Schrödinger Operators*, TX-LA Undergraduate Math Conference, Aug. 2020.

SOFTWARE

krtexas

R package accompanying *Nonparametric Regression via Tree-Guided Feature Aggregation*. Implements a penalized Nadaraya–Watson estimator with adaptive weights for joint nonparametric regression, feature aggregation, and variable selection over tree-structured predictors (e.g. microbiome data). Co-authored with Y. Samuel Wang and Martin T. Wells. GitHub.

permaBrioche

R package accompanying *Effect-size Estimation and Confounding-Aware Permutations for PERMANOVA*. Implements confounding-aware PERMANOVA and Hájek-based effect-size estimation in repeated-measures and other blocked microbiome study designs. Developed as part of the BioBakery ecosystem at the Huttenhower Lab, Harvard T.H. Chan School of Public Health. GitHub.

rPGBS

R package accompanying *Stable Variable Selection for High-Dimensional Genomic Data with Strong Correlations*. Implements two-stage hierarchical approach to variable selection consisting of a random pseudo-group clustering and bi-level variable selection Host.

TEACHING EXPERIENCE

Break Through Tech Artificial Intelligence (AI), Cornell Tech, Head TA (Summer 2026, Summer 2025, Summer 2024, Summer 2023)

Break Through Tech Artificial Intelligence (AI), Cornell Tech, Machine Learning Course Development Consultant (Spring 2024)

Introductory Statistics for Biology, Cornell University, Head TA (Fall 2023, Spring 2023, Fall 2022)

SERVICE

Cornell Statistics Graduate Society Student-Faculty Liaison (2023 - Present)

Cornell Advanced Graduate Teaching Cohort, Member (2022 - Present)

PROFESSIONAL MEMBERSHIPS

American Statistical Association (2021 - Present)